Jyotsna Pandey

jyotsnapandey6207@gmail.com jyotsnapandey-crypto Jyotsna Pandey

ABOUT ME

I am a Computer Science undergraduate with strong foundations in Java, Python, Data Structures, Algorithms, and Object-Oriented Programming. Experienced in developing full-stack applications using React.js, Flask, and Flutter, with hands-on experience in backend APIs, databases, and machine learning integration. Passionate about software engineering, scalable application development, and problem-solving.

EDUCATION

Bachelor of Technology 2027

VELLORE INSTITUTE OF TECHNOLOGY, Bhopal, Madhya Pradesh

CGPA: 8.75

Related Coursework: Computer Science and Engineering

Higher Secondary Examination 2022

DAV Public School Walmi Complex, Patna, Bihar

Percentage: 81.6

Major Subjects: Physics, Chemistry, Mathematics, and English

SKILLS

Technical Skills

- Languages:Java, Python, C++, JavaScript, SQL
- Core CS:Data Structures Algorithms, OOPs, DBMS
- Database Querying:SQL (SELECT, WHERE, GROUP BY, JOIN), JSON-based data handling C++ Concepts: STL, Pointers, Memory Management, OOP
- Software Development:Debugging, Code Optimization, Problem Solving
- Machine Learning: Scikit-learn, TensorFlow
- WebDevelopment: Proficient in HTML, CSS, JavaScript, React.js, and Tailwind CSS
- Tools: Git, GitHub, VS Code, Jupyter Notebook

Languages

- Hindi (Native)
- English (Fluent)
- Japanese (learning)

PROJECTS

Hydrottva

An app which helps in tracking the hydroponic plants , predict the yield percent and also estimate cost.

- Full-Stack Development: Engineered a full-stack mobile application using Flutter (frontend) and Flask (backend) to digitize hydroponic farm management, enabling real-time plant health monitoring and data-driven decision-making.
- Machine Learning and Computer Vision: Developed and integrated a TensorFlow/keras CNN model to analyze plant images , achieving 92 percent accuracy in identifying early-stage diseases and nutrient deficiencies, reducing potential crop loss by 15 percent.
- Improved performance by optimizing data processing logic and reducing computation time

- Predictive Analytics: Designed a yield prediction algorithm leveraging historical growth data and current health metrics, allowing farmers to forecast harvest yield with 85 percent confidence for improved supply chain and inventory planning.
- Cost Optimization: Built a profitability analysis tool that estimates operational costs and projects revenue based on predicted yield and market prices, identifying opportunities to reduce resource waste by 10 percent.

  Project Link: Hydrottva

Zaika Balance

Zaika Balance is a smart diet recommendation platform that personalizes Indian meal plans using AI. Built with React.js (frontend) and Flask (backend), it features nutrient tracking, calorie prediction, and personalized suggestions powered by machine learning models.

TECH- STACK

- Full-Stack Web Development: Architected and deployed a responsive full-stack web application using React.js (Vercel) and Flask (Render), creating a personalized platform for Indian diet management for over 1,000 users.
- Machine Learning Integration: Trained and deployed multiple scikit-learn and TensorFlow models (KNN, Regression) for personalized meal recommendations, nutrient tracking, and calorie prediction, leading to a 30 percent increase in user-reported dietary adherence.
- Performed debugging and optimization to enhance backend efficiency and response time
- Backend API and Data Management: Implemented a secure RESTful API with token-based JWT authentication and engineered a scalable data persistence layer using JSON to efficiently manage user profiles, meal data, and feedback.
- User-Centric Features: Developed core features including a nutrient dashboard, personalized weekly meal plans, and a feedback system, resulting in a 4.5/5 average user rating.

  Project Link: Zaika Balance

ACHIEVEMENTS AND EXPERIENCE

- Participated in university hackathon, building an IoT-based hydroponics solution
- Worked in a team-based development environment, contributing to frontend, backend, and ML modules
- Actively learned Machine Learning (NPTEL)

CERTIFICATIONS

HackerRank - python(Basic) Accomplishment certificate Certification Link
Vityarathi- For completing course of Python Essentials(Forage - April 2024) Certification Link
MATLAB Onramp-self-paced training course(Forage - Sept 2023) Certification Link